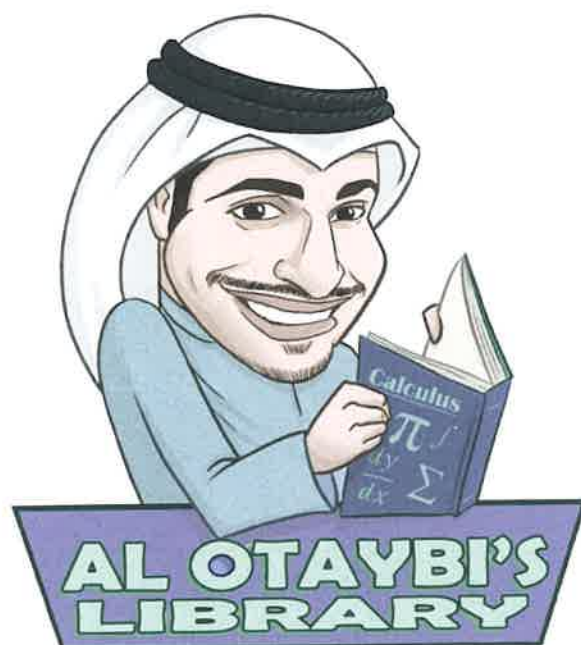


Kuwait University

2021/2022

Fall course



Calculus A

Second

Sections : 4.7 – 4.9 (Part a) and 5.2 (Part a)

4.7 Optimization Problems.

4.9 Antiderivatives.

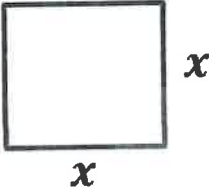
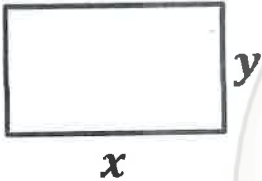
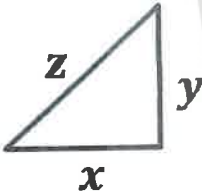
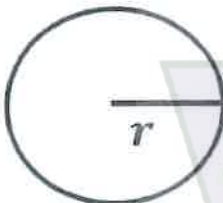
5.2 The Definite Integral.

(17)



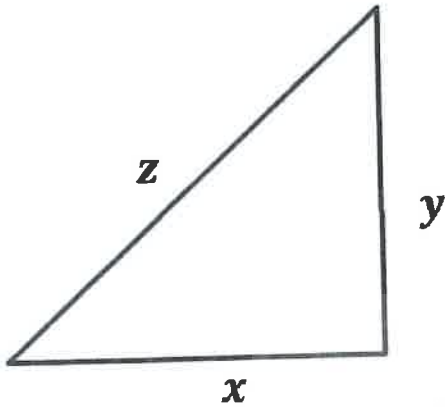
مراجعة وتذكير بالقوانين



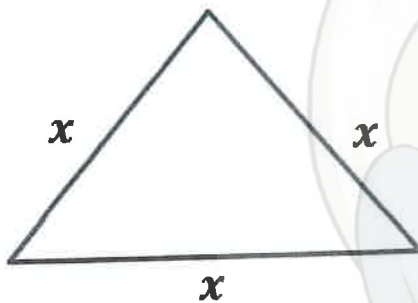
		Area المساحة	Perimeter or (Circumference) المحيط
	Square مربع	$A = x^2$	$C = 4x$
	Rectangle مستطيل	$A = x \cdot y$	$C = 2x + 2y$
	Right triangle مثلث قائم الزاوية	$A = \frac{1}{2} x \cdot y$	$C = x + y + z$
	Circle دائرة	$A = \pi \cdot r^2$	$A = 2 \cdot \pi \cdot r$



Special laws for Triangle



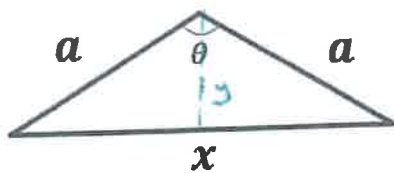
$$z^2 = x^2 + y^2$$



equilateral triangle
مثلث متساوي الأضلاع

$$A = \frac{\sqrt{3}}{4} x^2$$

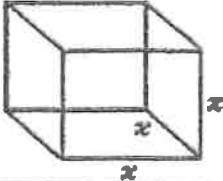
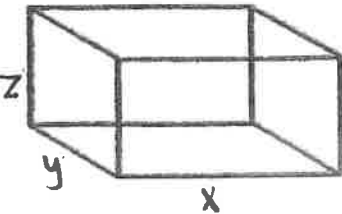
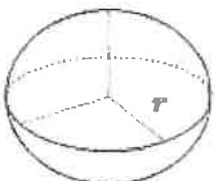
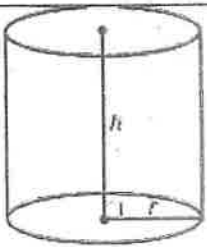
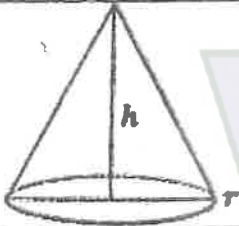
$$C = 3x$$



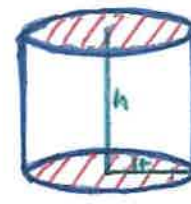
Isosceles triangle
مثلث متساوي الضلعين

$$A = \frac{1}{2} xy \quad \text{or} \quad A = \frac{1}{2} a^2 \sin \theta$$

$$C = 2a + x$$

		Volume (الحجم)	Surface Area (مساحة السطح)
	Cube (مكعب)	$V = x^3$	$S = 6x^2$
	Rectangular Box	$V = x \cdot y \cdot z$	$S = 2x \cdot y + 2x \cdot z + 2y \cdot z$
	Sphere (كرة) (ball)	$V = \frac{4}{3} \cdot \pi \cdot r^3$	$S = 4\pi \cdot r^2$
	Right Circular Cylinder (أسطوانة دائرية قائمة)	$V = \pi \cdot r^2 \cdot h$	$S = 2\pi \cdot r \cdot h$
	Cone (مخروط)	$V = \frac{1}{3} \cdot \pi \cdot r^2 \cdot h$	$S = \pi \cdot r \cdot \sqrt{r^2 + h^2}$

(اسطوانة دائرية قائمة) Right Circular Cylinder

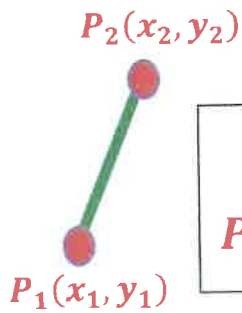
Open top and bottom	Open top (closed bottom)	Closed top and bottom
		
$S = 2\pi \cdot r \cdot h$	$S = 2\pi \cdot r \cdot h + \pi \cdot r^2$	$S = 2\pi \cdot r \cdot h + 2\pi \cdot r^2$
$V = \pi \cdot r^2 \cdot h$	$V = \pi \cdot r^2 \cdot h$	$V = \pi \cdot r^2 \cdot h$

(مخروط) cone

Open bottom	Closed bottom
	
$V = \frac{1}{3} \cdot \pi \cdot r^2 \cdot h$	$V = \frac{1}{3} \cdot \pi \cdot r^2 \cdot h$
$S = \pi \cdot r \cdot \sqrt{r^2 + h^2}$	$S = \pi \cdot r \cdot \sqrt{r^2 + h^2} + \pi \cdot r^2$



قوانين لا علاقة لها بالأشكال



Distance between point

$P_1(x_1, y_1)$ and $P_2(x_2, y_2)$

Distance by time

$$l = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

الزمن . السرعة = المسافة

(rate)

$$x = \frac{dx}{dt} \cdot t$$

$$l^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$



4.7 Optimization Problems



1) Domain

2) **المعادلة الأساسية**

المعادلة المساعدة

(3) وضع متغير بدلالة المتغير الآخر

4) $f'(x)$

$= 0$

undefind

أصفار بسط المشتقة

أصفار مقام المشتقة

ثم يتم التأكد من انتمائها للمجال

5) $f''(x)$

6) $f''(\text{الأعداد الناتجة})$

> 0

Minimum Value

< 0

Maximum Value





Text book question



Find two positive numbers whose product is 100 and whose sum is a minimum.

1) Domain = (0,100)

2) $S = x + y$ $x \cdot y = 100$

3) $x = \frac{100}{y}$

4) $S = \frac{100}{y} + y$

5) $S' = \frac{-100}{y^2} + 1$
 $= \frac{-100}{y^2} + \frac{y^2}{y^2}$
 $= \frac{-100 + y^2}{y^2}$

الممنوعات

(1) ممنوع أس سالب.

(2) ممنوع مقام غير موحد.

$= 0$

$-100 + y^2 = 0$

$y^2 = 100$

$y = 10 \in \text{Domain}$

$y = -10 \notin \text{Domain}$

undefind

$y = 0 \notin \text{Domain}$

6) $S'' = 200y^{-3} = \frac{200}{y^3}$

$S''(10) = \frac{200}{(10)^3} = \frac{200}{1000} = \frac{1}{5} > 0$

$\therefore y = 10 \xrightarrow{\text{من المساعدة}} x = \frac{100}{10} = 10$
 Give minimum Sum

Let x and y be two real numbers whose sum is 20.

Find the maximum values of their product.

Domain = $\mathbb{R}$

$$P = x \cdot y$$

$$x + y = 20$$

$$x = 20 - y$$

$$P = (20 - y) y$$

$$P = 20y - y^2$$

$$P' = 20 - 2y$$

$$= 0$$

$$20 - 2y = 0$$

$$20 = 2y$$

$$\div 2$$

$$y = 10 \in \text{Domain}$$

undefind
None

$$P'' = -2$$

$$P''(10) = -2 < 0$$

$$\therefore y = 10 \longrightarrow x = 20 - 10 = 10$$

give maximum P.

$$P_{\max} = 10(10) = 100$$



Text book question



Find two numbers whose difference is 100 and whose product is a minimum.

$$\text{Domain} = \mathbb{R}$$

$$P = x \cdot y$$

$$x - y = 100$$

$$x = 100 + y$$

$$P = (100 + y) \cdot y$$

$$P = 100y + y^2$$

$$P' = 100 + 2y$$

undefind
None

$$= 0$$

$$100 + 2y = 0$$

$$2y = -100$$

$$\div (2)$$

$$y = -50 \in \text{Domain}$$

$$P'' = 2$$

$$P''(-50) = 2 > 0$$

$$\therefore y = -50 \longrightarrow x = 100 + (-50) = 50$$

Give minimum Product.

Find the maximum product of two positive x and y
such that $2x + y = 100$

July 2018

$$\text{Domain} = (0, \infty)$$

$$P = x \cdot y$$

$$2x + y = 100$$

$$y = 100 - 2x$$

$$P = x(100 - 2x)$$

$$P = 100x - 2x^2$$

$$P' = 100 - 4x$$

$$= 0$$

undefind

None

$$100 - 4x = 0$$

$$4x = 100$$

$\div 4$

$$x = \frac{100}{4} = 25 \in \text{Domain}$$

$$P'' = -4$$

$$P''(25) = -4 < 0$$

$$\therefore x = 25 \longrightarrow y = 100 - 2(25) \\ = 100 - 50 \\ = 50$$

$$P_{\max} = 25(50) \\ = 1250$$

Given that $P = 2x^2 + 3y^2$ and $x - y = 10$.

Find x and y such that P is minimum.

Domain = $\mathbb{R}$

$$P = 2x^2 + 3y^2$$

$$x - y = 10$$

$$x = 10 + y$$

$$P = 2(10 + y)^2 + 3y^2$$

$$\begin{aligned} P' &= 4(10 + y) + 6y \\ &= 40 + 4y + 6y \\ &= 40 + 10y \end{aligned}$$

undefind
None

$$\begin{aligned} &= 0 \\ 40 + 10y &= 0 \end{aligned}$$

$$10y = -40$$

$\div 10$

$$y = -4 \in \text{Domain.}$$

$$P'' = 10$$

$$P''(-4) = 10 > 0$$

$$\therefore y = -4 \longrightarrow x = 10 + (-4) = 6$$

Give minimum $\triangleright$

Find the two positive numbers whose sum is 21, and the product of one of them times the square of the other is maximum.

$$\text{Domain} = (0, \infty)$$

$$P = x y^2$$

$$x + y = 21$$

$$x = 21 - y$$

$$P = (21 - y) y^2$$

$$P = 21 y^2 - y^3$$

$$P' = 42 y - 3 y^2$$

$$= 0$$

undefind

$$42 y - 3 y^2 = 0$$

$$y [42 - 3 y] = 0$$

$$y = 0$$

$$42 - 3 y = 0$$

⊄ Domain

$$42 = 3 y$$

÷3

$$y = \frac{42}{3} = 14 \in \text{Domain}$$

$$P'' = 42 - 6 y$$

$$P''(14) = 42 - 6(14) = 42 - 84 = -42 < 0$$

$$\therefore y = 14 \longrightarrow x = 21 - 14 = 7$$

give maximum P.



Text book question



The sum of two positive numbers is 16. What is the smallest possible value of the sum of their squares?

$$\text{Domain} = (0, \infty)$$

$$S = x^2 + y^2$$

$$x + y = 16$$

$$x = 16 - y$$

$$S = (16 - y)^2 + y^2$$

$$S' = 2(16 - y)(-1) + 2y$$

$$= -2(16 - y) + 2y$$

$$= -32 + 2y + 2y$$

$$= -32 + 4y$$

$$= 0$$

$$-32 + 4y = 0$$

$$4y = 32$$

$$\div 4$$

$$y = 8 \in \text{Domain}$$

$$S'' = 4$$

$$S''(8) = 4 > 0$$

$$y = 8$$

$$x = 16 - 8 = 8$$

Give minimum Sum.

$$S_{\min} = (8)^2 + (8)^2 = 64 + 64 = 128$$

Find the point on the curve $y = \sqrt{x}$ that is closest to the point $(2, 0)$

Jan 2017

Domain = $\mathbb{R}$

$$L^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$

$$L^2 = (x - 2)^2 + (y - 0)^2$$

$$L^2 = (x - 2)^2 + y^2$$

$$y = \sqrt{x}$$

$$L^2 = (x - 2)^2 + (\sqrt{x})^2$$

$$L^2 = (x - 2)^2 + x$$

$$(L^2)' = 2(x - 2) + 1$$

$$= 2x - 4 + 1$$

$$= 2x - 3$$

$$= 0$$

$$2x - 3 = 0$$

$$2x = 3$$

$\div 2$

$$x = \frac{3}{2} \in \text{Domain}$$

undefind

None

$$(L^2)'' = 2$$

$$(L^2)''\left(\frac{3}{2}\right) = 2 > 0$$

$$\therefore x = \frac{3}{2} \longrightarrow y = \sqrt{\frac{3}{2}}$$

$\therefore \left(\frac{3}{2}, \sqrt{\frac{3}{2}}\right)$ is closest to the point $(2, 0)$

Find the points on the graph of $\frac{x^2}{4} - y^2 = 1$
that are closest to the point $P(5, 0)$

Domain = $\mathbb{R}$

$$L^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$

$$L^2 = (x_1 - 5)^2 + (y_1 - 0)^2$$

$$L^2 = (x - 5)^2 + y^2$$

$$\frac{x^2}{4} - y^2 = 1$$

$$y^2 = \frac{x^2}{4} - 1$$

$$L^2 = (x - 5)^2 + \frac{x^2}{4} - 1$$

$$(L^2)' = 2(x - 5) + \frac{2x}{4}$$

$$= 2x - 10 + \frac{x}{2}$$

$$= \frac{4x - 20 + x}{2}$$

$$= \frac{5x - 20}{2}$$

$$= 0$$

$$5x - 20 = 0$$

$$5x = 20$$

$\div 5$

$$x = 4 \in \text{Domain}$$

undefined

$$2 = 0$$

None

$$(L^2)'' = 2 - 0 + \frac{1}{2} = 2 + \frac{1}{2} > 0$$

$$\text{at } x = 4 \longrightarrow y^2 = \frac{16}{4} - 1 = 4 - 1 = 3$$

$$y = \pm \sqrt{3}$$

$\therefore (4, \pm\sqrt{3})$ are closest to P (Minimum)

Find the point on the curve $y^2 = 2x$ that is closest to the point $(1, 4)$.

Domain = $\mathbb{R}$

$$L^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$L^2 = (x_2 - 1)^2 + (y_2 - 4)^2$$

$$L^2 = \left(\frac{y^2}{2} - 1\right)^2 + (y_2 - 4)^2$$

$$L^2 = \left(\frac{y^2}{2} - 1\right)^2 + (y - 4)^2$$

$$(L^2)' = 2\left(\frac{y^2}{2} - 1\right) \frac{2y}{2} + 2(y - 4)$$

$$= 2y\left(\frac{y^2}{2} - 1\right) + 2y - 8$$

$$= y^3 - 2y + 2y - 8$$

$$= y^3 - 8$$

$$= 0 \quad \text{DNE None}$$

$$y^3 - 8 = 0$$

$$y^3 = 8$$

$$y = 2$$

$\in \text{Domain}$

$$(L^2)'' = 3y^2$$

at $y = 2$

$$(L^2)'' = 3(4) = 12 > 0$$

$$y^2 = 2x$$

$$x = \frac{y^2}{2}$$



$\therefore y = 2 \rightarrow x = \frac{4}{2} = 2$
 Give minimum L^2
 (2, 2)

If the sum of the radius and height of a cylinder is 3 m
Find the dimensions of the cylinder that
maximize its volume.

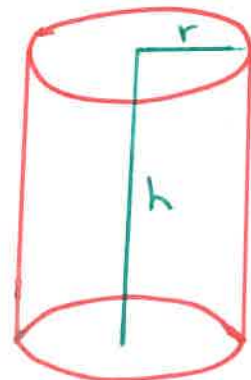
May 2018

$$\text{Domain} = (0, \infty)$$

$$V = \pi r^2 h$$

$$r + h = 3$$

$$h = 3 - r$$



$$V = \pi r^2 (3 - r)$$

$$V = 3\pi r^2 - \pi r^3$$

$$V' = 6\pi r - 3\pi r^2$$

$$= 0$$

undefind
None

$$6\pi r - 3\pi r^2 = 0$$

$$r [6\pi - 3\pi r] = 0$$

$$r = 0 \notin \text{Domain}$$

$$6\pi - 3\pi r = 0$$

$$3\pi r = 6\pi$$

$$\div 3\pi$$

$$r = \frac{6\pi}{3\pi} = 2 \in \text{Domain}$$

$$V'' = 6\pi - 6\pi r$$

$$V''(2) = 6\pi - 12\pi = -6\pi < 0$$

$$r = 2 \longrightarrow h = 3 - 2 = 1$$

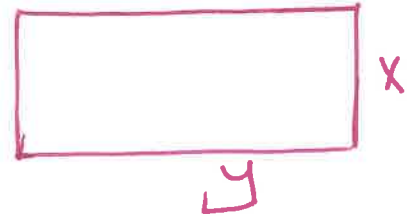
give maximum V

Find the dimension of a rectangle with perimeter 120 meter whose area is maximum

$$\text{Domain} = (0, 120)$$

$$A = x \cdot y$$

$$P = 2x + 2y$$



$$A = x(60 - x) \\ = 60x - x^2$$

$$120 = 2x + 2y$$

$$2y = 120 - 2x$$

$$y = 60 - x$$

$$A' = 60 - 2x$$

$$= 0$$

DNE
NONE

$$60 - 2x = 0$$

$$2x = 60$$

$$x = 30$$

∈ Domain

$$A'' = -2$$

$$\text{at } x = 30$$

$$A'' = -2 < 0$$

∴ at $x = 30 \text{ m} \rightarrow y = 60 - 30 = 30 \text{ m}$
give maximum A

$$A_{\max} = 30(30) = 900 \text{ m}^2$$

لوقال
Find maximum Area

Find the dimension of rectangle of maximum area
whose diagonal is 2 ft.

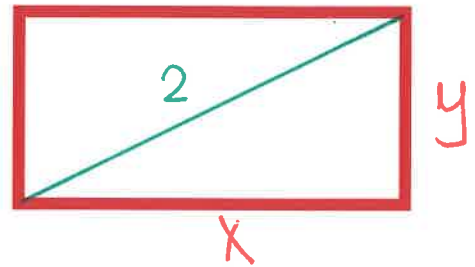
Domain = (0, 2)

$$A = x \cdot y$$

$$x^2 + y^2 = 4$$

$$x^2 = 4 - y^2$$

$$x = \sqrt{4 - y^2}$$



$$A = \sqrt{4 - y^2} \cdot y$$

$$A' = \frac{-2y}{2\sqrt{4 - y^2}} y + \sqrt{4 - y^2} \quad (1)$$

$$= \frac{-y^2}{\sqrt{4 - y^2}} + \frac{\sqrt{4 - y^2} \sqrt{4 - y^2}}{\sqrt{4 - y^2}}$$

$$= \frac{-y^2 + 4 - y^2}{\sqrt{4 - y^2}}$$

$$= \frac{4 - 2y^2}{\sqrt{4 - y^2}}$$

$$= 0$$

$$4 - 2y^2 = 0$$

$$2y^2 = 4$$

$$\div 2 \quad y^2 = 2$$

$$y = -\sqrt{2} \notin \text{Domain}, y = \sqrt{2} \in \text{Domain.}$$

undefind

$$4 - y^2 = 0$$

$$y^2 = 4$$

$$y = \pm 2 \notin \text{Domain}$$

$$A'' = \frac{\sqrt{4-y^2}(-4y) - (4-2y^2) \frac{-2y}{2\sqrt{4-y^2}}}{4-y^2}$$

$$\begin{aligned} A''(\sqrt{2}) &= \frac{\sqrt{4-2}(-4\sqrt{2}) - 0}{4-2} \\ &= \frac{\sqrt{2}(-4\sqrt{2})}{2} \\ &= \frac{-4(2)}{2} = -4 < 0 \end{aligned}$$

$$\therefore y = \sqrt{2} \longrightarrow x = \sqrt{4-2} = \sqrt{2}$$

gives maximum A

AL OTAYBI'S
LIBRARY

A closed box of square base is to be made to have a volume of $64ft^3$. If the cost of the material used is 50 cent per ft and if there is no waste of material. Find the dimension that will minimize the cost of material.

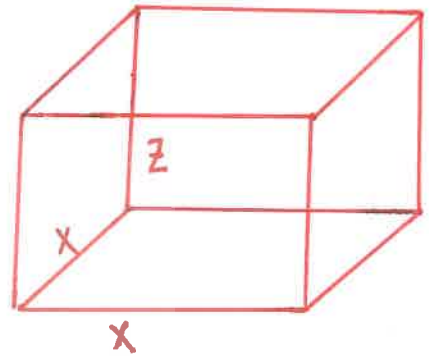
Domain = $(0, \infty)$

$C = (4xz + 2x^2) 50$

$V = 64$

$x^2z = 64$

$z = \frac{64}{x^2}$



$C = (4x \frac{64}{x^2} + 2x^2) 50$

$C' = (\frac{-256}{x^2} + 4x) 50$

$= 50 \left(\frac{-256 + 4x^3}{x^2} \right)$

$= 0$ ← undefind

$x = 0 \notin \text{Domain.}$

$50(-256 + 4x^3) = 0$

$\div 50$

$-256 + 4x^3 = 0$

$\div 4$

$4x^3 = 256$

$x^3 = \frac{256}{4} = 64$

$x = 4 \in \text{Domain}$

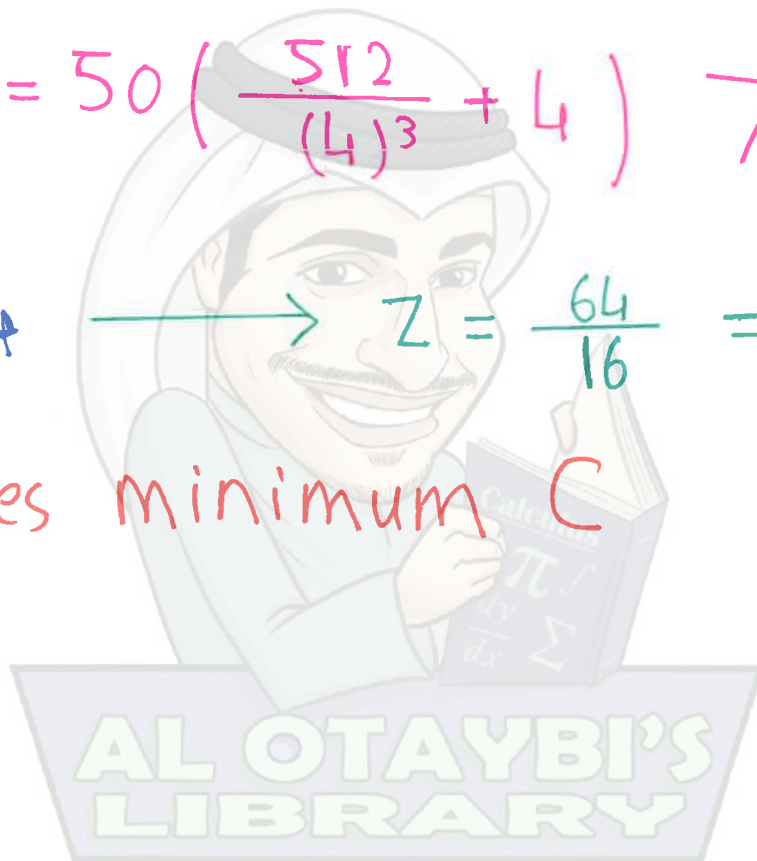
$$C' = (-256x^{-2} + 4x) \quad \text{So}$$

$$\begin{aligned} C'' &= 50(-256(-2)x^{-3} + 4) \\ &= 50\left(\frac{512}{x^3} + 4\right) \end{aligned}$$

$$C''(4) = 50\left(\frac{512}{(4)^3} + 4\right) > 0$$

$$x = 4 \longrightarrow z = \frac{64}{16} = 4$$

gives minimum C



A rectangular box with a square base and open top has a volume $4m^3$. Find the dimension of the box that minimize the surface area.

$$\text{Domain} = (0, \infty)$$

$$S = 4xz + x^2$$

$$S = 4x \frac{4}{x^2} + x^2$$

$$S = \frac{16}{x} + x^2$$

$$S' = -\frac{16}{x^2} + 2x$$

$$= -\frac{16}{x^2} + \frac{2x \cdot x^2}{x^2}$$

$$= \frac{-16 + 2x^3}{x^2}$$

$$= 0$$

$$-16 + 2x^3 = 0$$

$$2x^3 = 16$$

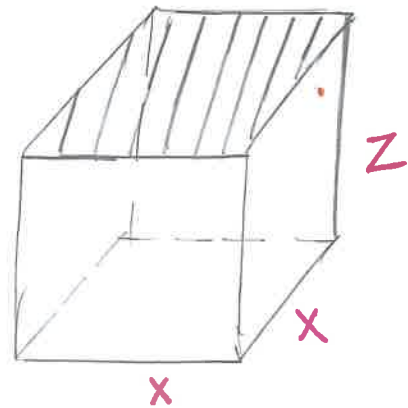
$$x^3 = 8$$

$$x = 2$$

$\in \text{Domain}$

$$zx^2 = 4$$

$$z = \frac{4}{x^2}$$



#

DNE (undefined)

$$x = 0$$

$\notin \text{Domain}$

$$S'' = \frac{x^2 \overset{6}{0} - (-16) 2x}{x^4} + 2$$

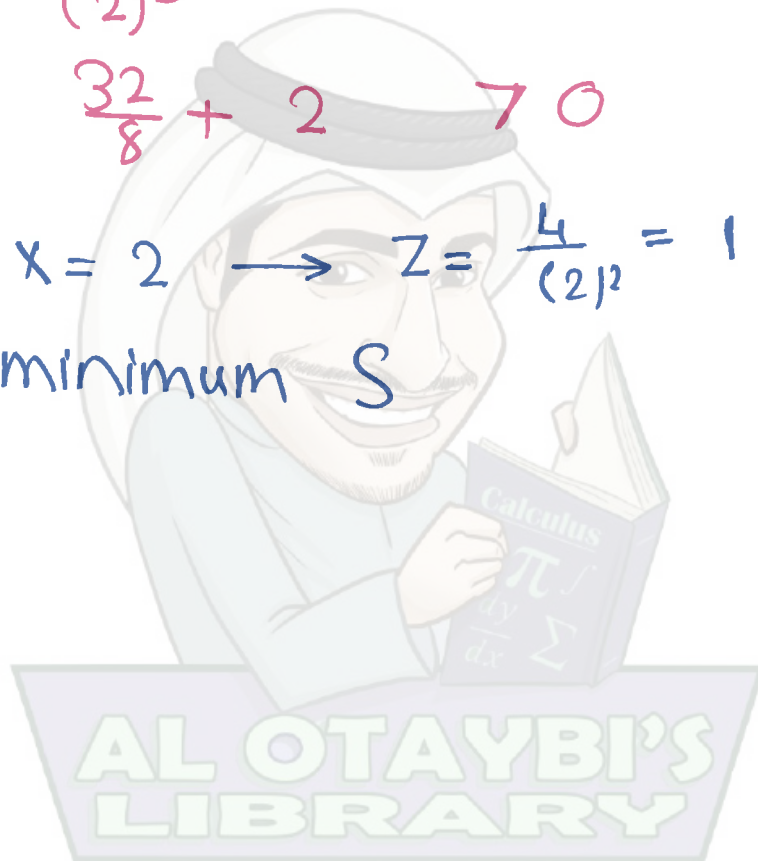
$$= \frac{32}{x^3} + 2$$

at $x = 2$

$$S'' = \frac{32}{(2)^3} + 2$$

$$= \frac{32}{8} + 2 = 7 > 0$$

$\therefore$ at $x = 2 \rightarrow Z = \frac{4}{(2)^2} = 1$
 give minimum S



A farmer has 200 *meters* of fence to be used in constructing three sides of a rectangle. An existing long straight wall is to be used for the fourth side. What dimensions will maximize the area of the rectangle ?

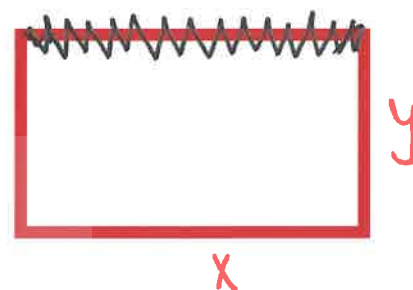
$$\text{Domain} = (0, 200)$$

$$A = x \cdot y$$

$$y + x + y = 200$$

$$2y + x = 200$$

$$x = 200 - 2y$$



$$A = (200 - 2y) y$$

$$A = 200y - 2y^2$$

$$A' = 200 - 4y$$

$$= 0$$

$$200 - 4y = 0$$

$$200 = 4y$$

$\div 4$

$$y = \frac{200}{4} = 50 \in \text{Domain}$$

undefind

None

$$A'' = -4$$

$$A''(50) = -4 < 0$$

$$\therefore y = 50 \longrightarrow x = 200 - 2(50) = 200 - 100$$

$$x = 100$$

gives maximum A

إذا كانت المعادلة الأساسية مكونة من متغير واحد فقط



يتم إهمال الخطوة

2 - 3 - 4



Let A be the area of a square, and P is perimeter.

Find the minimum value of $y = A - P$

Domain = $(0, \infty)$

$$y = A - P$$

$$y = x^2 - 4x$$

$$y' = 2x - 4$$

$$= 0$$

$$2x - 4 = 0$$

$$2x = 4$$

$$\div 2$$

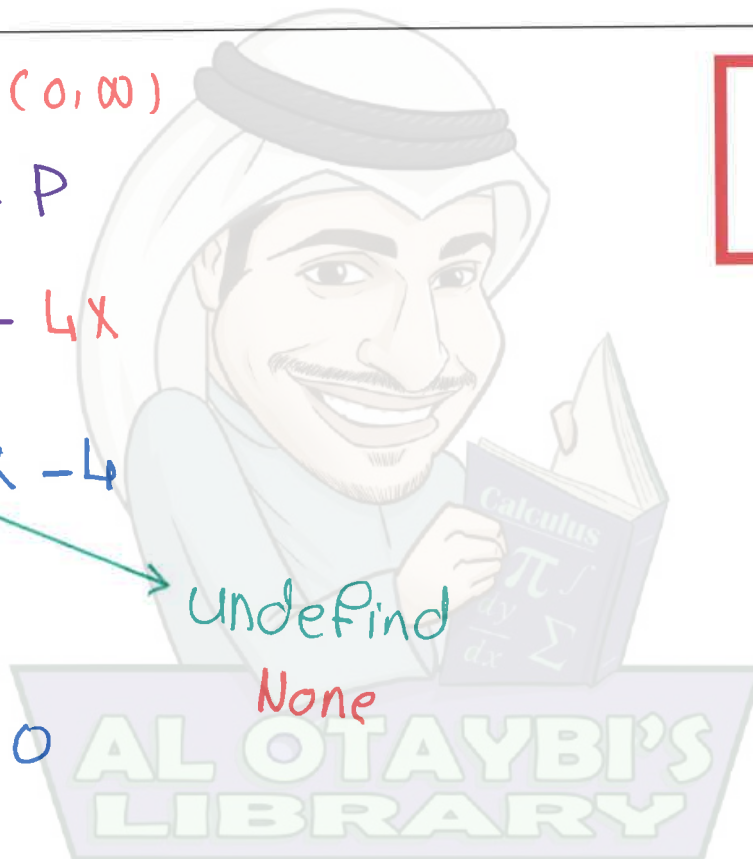
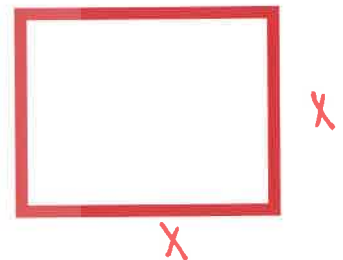
$$x = 2 \in \text{Domain}$$

$$y'' = 2$$

$$y''(2) = 2 > 0$$

→ $x = 2$ gives minimum y

$$y_{\min} = (2)^2 - 4(2) = 4 - 8 = -4$$



AL OTAYBI'S
LIBRARY

Find the minimum value of $S = 9x + 8y^2$ where $x(y^2 + 1) = 2$

$$\text{Domain} = \mathbb{R}$$

$$S = 9x + 8y^2$$

$$S = 9x + 8\left(\frac{2}{x} - 1\right)$$

$$S = 9x + \frac{16}{x} - 8$$

$$S' = 9 - \frac{16}{x^2} \quad \#$$

$$S' = \frac{9x^2}{x^2} - \frac{16}{x^2}$$
$$= \frac{9x^2 - 16}{x^2}$$

$= 0$

DNE (undefined)

$$9x^2 - 16 = 0$$

$$9x^2 = 16$$

$$x^2 = \frac{16}{9}$$

$$x = \pm \sqrt{\frac{16}{9}}$$

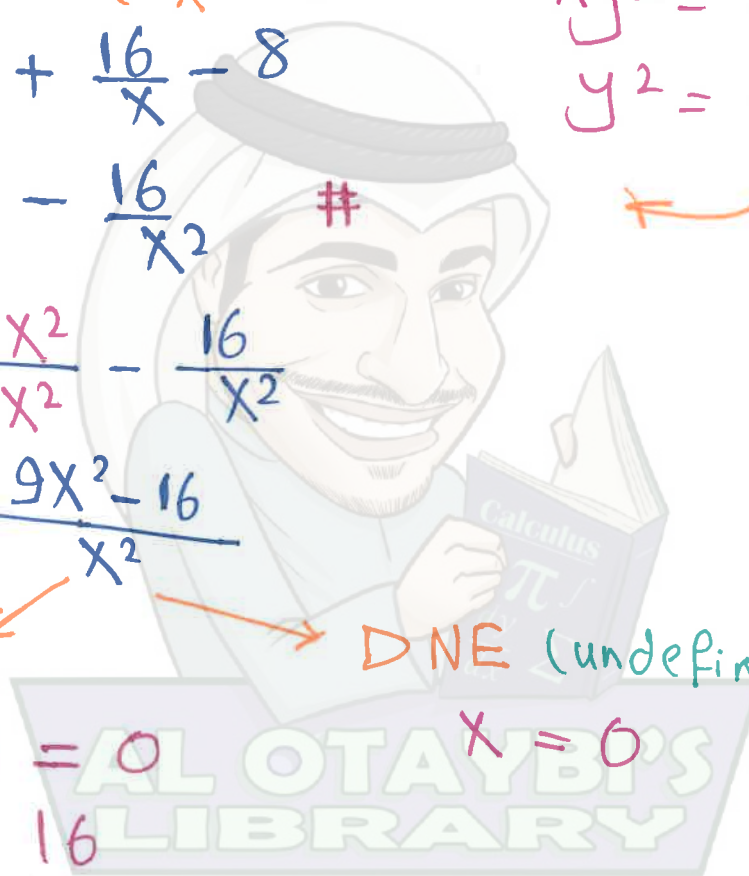
$$x = \frac{4}{3} \quad x = -\frac{4}{3}$$

$$x(y^2 + 1) = 2$$

$$xy^2 + x = 2$$

$$xy^2 = 2 - x$$

$$y^2 = \frac{2}{x} - 1$$



$$\begin{aligned}
 S'' &= 0 - \frac{x^2(0) - 16 \cdot 2x}{x^4} \\
 &= - \frac{-32x}{x^4} \\
 &= \frac{32}{x^3}
 \end{aligned}$$

at $x=0 \rightarrow S'' \text{ DNE}$

at $x = \frac{4}{3} \rightarrow S'' = \frac{32}{(\frac{4}{3})^3} > 0$

at $x = -\frac{4}{3} \rightarrow S'' = \frac{32}{(-\frac{4}{3})^3} < 0$

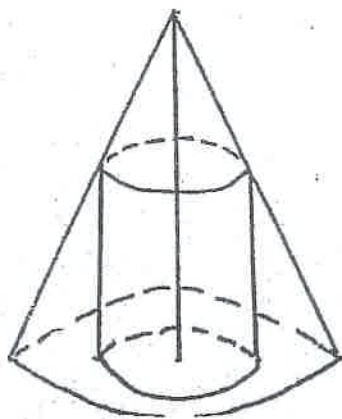
$\therefore$ at $x = \frac{4}{3} \rightarrow y^2 = \frac{2}{\frac{4}{3}} - 1$

give minimum $S = \frac{6}{4} - 1$

$$\begin{aligned}
 &= \frac{3}{2} - 1 \\
 &= \frac{3}{2} - \frac{2}{2} = \frac{1}{2} \\
 y &= \pm \frac{1}{\sqrt{2}}
 \end{aligned}$$

$$\begin{aligned}
 S_{\text{minimum}} &= 9x + 8y^2 \\
 &= 9\left(\frac{4}{3}\right) + 8\left(\frac{1}{2}\right) \\
 &= 3(4) + 8\left(\frac{1}{2}\right) = 12 + 4 = 16
 \end{aligned}$$

11) Find the dimension of the hollow cylinder of maximum surface area that can be inscribed in a cone of altitude 10 cm and base radius 5 cm, if the axes of the cylinder and cone coincide.



Domain = (0,10)

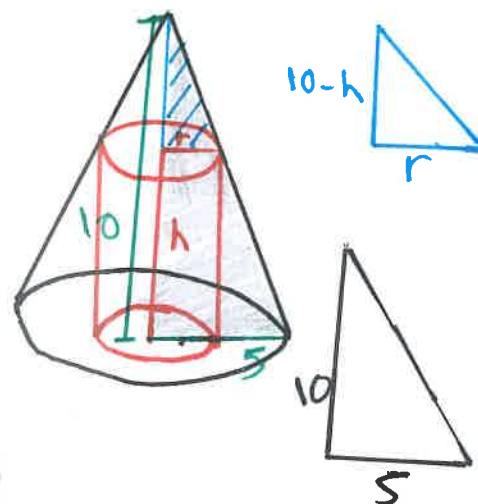
$$S = 2\pi rh$$

$$\frac{10-h}{r} = \frac{10}{5}$$

$$\frac{10-h}{r} = 2$$

$$10-h = 2r$$

$$h = 10 - 2r$$



$$S = 2\pi r(10-2r)$$

$$S = 20\pi r - 4\pi r^2$$

$$S' = 20\pi - 8\pi r$$

= 0

$$20\pi - 8\pi r = 0$$

$$8\pi r = 20\pi$$

$\div 8\pi$

$$r = \frac{20\pi}{8\pi} = \frac{5}{2} \in \text{Domain}$$

$$S'' = -8\pi$$

$$S''\left(\frac{5}{2}\right) = -8\pi$$

$$\rightarrow \therefore r = \frac{5}{2} \rightarrow h = 10 - 2\left(\frac{5}{2}\right) = 5$$

gives maximum S

Find the maximum area of the rectangle that can be inscribed inside a semi circle of radius 2.

Domain : (0,2)

عين + طول

$$A = (2x) y$$

$$A = 2x \sqrt{4 - x^2}$$

الشتقا بشا + بشا اشتقا

$$A' = 2(1)\sqrt{4-x^2} + 2x \frac{-2x}{2\sqrt{4-x^2}}$$

$$= 2\sqrt{4-x^2} - \frac{2x^2}{\sqrt{4-x^2}}$$

$$= \frac{2\sqrt{4-x^2}\sqrt{4-x^2}}{\sqrt{4-x^2}} - \frac{2x^2}{\sqrt{4-x^2}}$$

$$= \frac{2(4-x^2) - 2x^2}{\sqrt{4-x^2}} = \frac{8-2x^2-2x^2}{\sqrt{4-x^2}}$$

$$= \frac{8-4x^2}{\sqrt{4-x^2}}$$

$$= 0$$

$$8 - 4x^2 = 0$$

$$\div 4 \quad 8 = 4x^2$$

$$x^2 = 2$$

$$x = \sqrt{2}$$

∈ Domain

$$x = -\sqrt{2}$$

∉ Domain

D.N.E

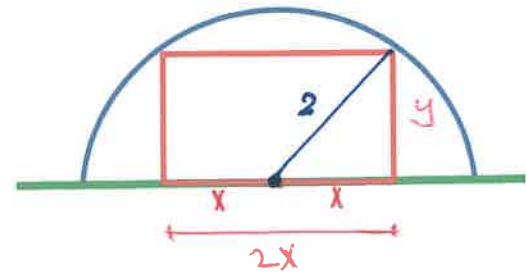
$$4 - x^2 = 0$$

$$4 = x^2$$

$$x = -2 \quad x = 2$$

∉ Domain

∉ Domain



فيثاغورث

$$(2)^2 = x^2 + y^2$$

$$4 = x^2 + y^2$$

$$y^2 = 4 - x^2$$

$$y = \sqrt{4 - x^2}$$

$$A'' = \frac{\sqrt{4-x^2} (-8x) - (8-4x^2) \frac{-2x}{2\sqrt{4-x^2}}}{4-x^2}$$

at $x = \sqrt{2}$

$$A'' = \frac{\sqrt{4-(\sqrt{2})^2} (-8)\sqrt{2} - 0}{4 - (\sqrt{2})^2}$$

$$= \frac{\sqrt{4-2} (-8)\sqrt{2}}{4-2}$$

$$= \frac{\sqrt{2} (-8)\sqrt{2}}{2} = \frac{2(-8)}{2} = -8 < 0$$

$\therefore x = \sqrt{2}$ $\rightarrow y = \sqrt{4-(\sqrt{2})^2} = \sqrt{4-2} = \sqrt{2}$
 give maximum A

$$A_{\max} = 2xy = 2\sqrt{2}\sqrt{2} = 2(2) = 4$$



4.9 Antiderivatives



$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$



Examples

$$\int x^3 dx = \frac{x^4}{4} + c$$

$$\int x^4 dx = \frac{x^5}{5} + c$$

$$\int x^{-3} dx = \frac{x^{-2}}{-2} + c$$

$$\int x^{-4} dx = \frac{x^{-3}}{-3} + c$$

$$\int x^{\frac{4}{3}} dx = \frac{x^{\frac{7}{3}}}{\frac{7}{3}} + c$$

$$\int x^{-\frac{3}{2}} dx = \frac{x^{-\frac{1}{2}}}{-\frac{1}{2}} + c$$

$$\int (x^3 + x^7 - x^{-2} + x^{-4} - x) dx$$

يمكن توزيع التكامل
على الجمع والطرح فقط

$$= \frac{x^4}{4} + \frac{x^8}{8} - \frac{x^{-1}}{-1} + \frac{x^{-3}}{-3} - \frac{x^2}{2} + c$$



تكامل الرقم



بروحه



أعطيه ايس

بجانبه اكسات



ينزل الرقم كمال هو
ثم أكامل الايس

Examples

$$\int 5 dx = 5x + c$$

$$\int 5x^7 dx = 5 \frac{x^8}{8} + c$$

$$\int c f(x) dx = c \int f(x) dx$$



ملاحظات حلوة



Thickness
dx

كل تكامل يحتوي على

constant
c

كل تكامل غير محدد ناتجه يحتوي على



ممنوعاً

ممنوع إجراء التكامل على

ضرب

قسمة

أس مرفوع

نضرب
ثم نكامل

نقسم
ثم نكامل

نبطل الأس
ثم نكامل

Examples

$$\int (x^3 + 3)^2 dx$$

$$= \int [x^6 + 6x^3 + 9] dx$$

$$= \frac{x^7}{7} + 6 \frac{x^4}{4} + 9x + C$$

$$\int \frac{x^2 + 1}{x^2} dx$$

$$= \int \left[\frac{x^2}{x^2} + \frac{1}{x^2} \right] dx$$

$$= \int [1 + x^{-2}] dx$$

$$= x + \frac{x^{-1}}{-1} + C$$

$$\int x^3(x^2 + 2x) dx$$

$$= \int (x^5 + 2x^4) dx$$

$$= \frac{x^6}{6} + 2 \frac{x^5}{5} + C$$

$$\int (x^2 + 2x + \frac{5}{x^2} - 2) dx$$

$$= \int (x^2 + 2x + 5x^{-2} - 2) dx$$

قوة

$$= \frac{x^3}{3} + 2 \frac{x^2}{2} + 5 \frac{x^{-1}}{-1} - 2x + C$$

$$\int (\pi^2 + e - \pi x) dx$$

ثابت

$$= \pi^2 x + ex - \pi \frac{x^2}{2} + C$$

$$\int \sqrt{x}(\sqrt{x} + \sqrt{3}) dx$$

$$= \int (x + \sqrt{3} \sqrt{x}) dx$$

$$= \int (x + \sqrt{3} x^{\frac{1}{2}}) dx$$

$$= \frac{x^2}{2} + \sqrt{3} \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$\int \left(x - \frac{1}{x}\right)^2 dx$$

$$= \int (x^2 - 2x \cdot \frac{1}{x} + \frac{1}{x^2}) dx$$

$$= \int (x^2 - 2 + x^{-2}) dx$$

$$= \frac{x^3}{3} - 2x + \frac{x^{-1}}{-1} + C$$

$$\int \frac{(\sqrt{t} + 2)^2}{t^3} dt$$

$$= \int \frac{t + 4\sqrt{t} + 4}{t^3} dt$$

$$= \int \left(\frac{t}{t^3} + 4 \frac{t^{\frac{1}{2}}}{t^3} + \frac{4}{t^3} \right) dt$$

$$= \int \left(\frac{1}{t^2} + 4t^{\frac{1}{2}-3} + \frac{4}{t^3} \right) dt$$

$$= \int (t^{-2} + 4t^{-\frac{5}{2}} + 4t^{-3}) dt$$

$\frac{1}{2} - 3$
 $\frac{1}{2} - \frac{6}{2}$
 $-\frac{5}{2}$

$$= \frac{t^{-1}}{-1} + 4 \frac{t^{-\frac{3}{2}}}{-\frac{3}{2}} + 4 \frac{t^{-2}}{-2} + C$$



Tricky question !!



$$\int \frac{x^3 + 3x^2 - 9x - 2}{x - 2} dx$$

$$\begin{array}{r} x^2 + 5x + 1 \\ -2 + x \overline{) x^3 + 3x^2 - 9x - 2} \\ \underline{-(x^3 + 2x^2)} \\ 5x^2 - 9x - 2 \\ \underline{-(5x^2 + 10x)} \\ x - 2 \\ \underline{-(x - 2)} \\ 0 \end{array}$$

$\int (x^2 + 5x + 1) dx$

$$= \frac{x^3}{3} + 5 \frac{x^2}{2} + x + C$$



Text book questions



Find the most general antiderivatives of the function

$$f(x) = 3\sqrt{x} - 2\sqrt[3]{x}$$

$$\begin{aligned} & \int (3\sqrt{x} - 2\sqrt[3]{x}) dx \\ &= \int (3x^{\frac{1}{2}} - 2x^{\frac{1}{3}}) dx \\ &= 3 \frac{x^{\frac{3}{2}}}{\frac{3}{2}} - 2 \frac{x^{\frac{4}{3}}}{\frac{4}{3}} + C \end{aligned}$$

Find the most general antiderivatives of the function

$$g(t) = \frac{1 + t + t^2}{\sqrt{t}}$$

$$\begin{aligned} & \int \frac{1 + t + t^2}{t^{\frac{1}{2}}} dt \\ &= \int \left(\frac{1}{t^{\frac{1}{2}}} + \frac{t}{t^{\frac{1}{2}}} + \frac{t^2}{t^{\frac{1}{2}}} \right) dt \\ &= \int \left(t^{-\frac{1}{2}} + t^{1-\frac{1}{2}} + t^{2-\frac{1}{2}} \right) dt \quad \begin{array}{l} \xrightarrow{2-\frac{1}{2}=\frac{3}{2}} \\ \xrightarrow{\frac{1}{2}-\frac{1}{2}=\frac{0}{2}} \end{array} \\ &= \int \left(t^{-\frac{1}{2}} + t^{\frac{1}{2}} + t^{\frac{3}{2}} \right) dt \\ &= \frac{t^{\frac{1}{2}}}{\frac{1}{2}} + \frac{t^{\frac{3}{2}}}{\frac{3}{2}} + \frac{t^{\frac{5}{2}}}{\frac{5}{2}} + C \end{aligned}$$



Text book question



Find the most general antiderivatives of the function
(Check your answer by differentiation)

$$f(x) = x(12x + 8)$$

$$\begin{aligned} & \int x(12x + 8) dx \\ &= \int (12x^2 + 8x) dx \\ &= 12 \frac{x^3}{3} + 8 \frac{x^2}{2} + C \\ &= 4x^3 + 4x^2 + C \end{aligned}$$

Check

$$\begin{aligned} & \frac{d}{dx} (4x^3 + 4x^2 + C) \\ &= 12x^2 + 8x + 0 \\ &= 12x^2 + 8x \\ &= x(12x + 8) = f(x) \end{aligned}$$



Definite integral

التكامل المحدد



لا يوجد **C**
والناتج النهائي عدد



$$\int_0^1 x \, dx$$

$$= \left[\frac{x^2}{2} \right]_0^1 = \frac{(1)^2}{2} - \frac{(0)^2}{2} = \frac{1}{2} - 0 = \frac{1}{2}$$

$$\int_4^9 \frac{1}{\sqrt{x}} \, dx$$

$$\begin{aligned} &= \int_4^9 x^{-\frac{1}{2}} \, dx \\ &= \left[\frac{x^{\frac{1}{2}}}{\frac{1}{2}} \right]_4^9 = \left[2x^{\frac{1}{2}} \right]_4^9 \\ &= 2(9)^{\frac{1}{2}} - 2(4)^{\frac{1}{2}} \\ &= 2(3) - 2(2) = 6 - 4 = 2 \end{aligned}$$

Evaluate the following integral

$$\int_{-2}^3 (x^2 - 3) dx$$

$$= \left[\frac{x^3}{3} - 3x \right]_{-2}^3$$

$$= \left[\frac{27}{3} - 9 \right] - \left[\frac{-8}{3} + 6 \right]$$

$$= 9 - 9 - \left[\frac{-8}{3} + \frac{18}{3} \right]$$

$$= 0 - \left[\frac{10}{3} \right] = -\frac{10}{3}$$

$$\int_0^{\pi} (5e^x + 3\sin x) dx$$

$$= \left[5e^x - 3\cos x \right]_0^{\pi}$$

$$= \left[5e^{\pi} - 3\cos \pi \right] - \left[5e^0 - 3\cos 0 \right]$$

$$= \left[5e^{\pi} - 3(-1) \right] - \left[5 - 3 \right]$$

$$= 5e^{\pi} + 3 - 2$$

$$= 5e^{\pi} + 1$$



هناك شرط لاجراء التكامل المحدد وهو
اتصال الدالة بفتره حدود التكامل



Can you evaluate

$$\int_0^3 \frac{x^3}{x-1} dx ??$$

$$f(x) = \frac{x^3}{x-1}$$

Domain of $f(x) = \mathbb{R} \setminus \{1\}$

$$\therefore 1 \in [0, 3]$$

$\therefore$ We can Not evaluate the integral.

Can you evaluate

$$\int_0^3 \frac{1}{x+1} dx ??$$

$$f(x) = \frac{1}{x+1}$$

Domain of $f(x) = \mathbb{R} \setminus \{-1\}$

$$\therefore -1 \notin [0, 3]$$

$\therefore$ We Can Evaluate the integral.

Can you evaluate $\int_0^1 \frac{2x^2 - x - 1}{2x + 1} dx$??

If yes, justify your answer and find the value of the integral.

$$f(x) = \frac{2x^2 - x - 1}{2x + 1}$$

Domain of $f(x)$: $2x + 1 = 0$
 $2x = -1$
 $x = -\frac{1}{2}$

$$\Rightarrow D_f = \mathbb{R} \setminus \left\{ -\frac{1}{2} \right\}$$

$\therefore -\frac{1}{2} \notin [0, 1] \Rightarrow \therefore$ We Can Evaluate the integral.

$$\int_0^1 \frac{2x^2 - x - 1}{2x + 1} dx$$

$$\int_0^1 \frac{(2x + 1)(x - 1)}{2x + 1} dx$$

$$\int_0^1 (x - 1) dx$$

$$\left[\frac{x^2}{2} - x \right]_0^1 = \left(\frac{1^2}{2} - 1 \right) - 0$$
$$= \frac{1}{2} - 1 = -\frac{1}{2}$$